
Active Directory Lab

Objectives

- Create a VM in Azure
- Promote a windows Server to a Domain controller
- Create OU, Users, and security groups in two different ways. (PowerShell and Server Manger)
- Implement and enforce group policy in the security group
- show popular help desk tasks, for example password reset.
- Lastly Create a second VM and connect it to the domain.

Business problem This Lab Solves

This lab will help a Business organize their employees and what they have access to. An Active Directory does it all. It keeps all users' information in a secure place where only those who are authorized can access the info easily.

VM Setting

Cloud Platform	Microsoft Azure
VM Size	At least 2cpu and 4gb Ram (Standard D2 and B2 is cheapest option)
OS	Windows Server 2025 Datacenter Gen2
Domain	Test.Forest
Costs	Less than \$10

Why Azure

Really you can use any cloud platform to do this project (AWS, Azure, or GCP) since they are the most cost efficient do this lab. Cloud platforms usually use a pay as go service which means you are not paying for anything you are not using unlike using your own data center and it is easier to sale up or down. Now the reason I'm using azure is because it's simply naming convention and easy learning path.

What was Built

Domain Controller

- Install Active Directory Domain server in our server manager
- Promote window server into a domain controller
- Create a Forest Test.Forest
- Set a disaster recovery password in case any thing happens

OU's and user

OU	Security Group	Users	How they were created
IT	IT Group	Demarcus Miller	Server Manger
HR	HR Group	Magic Miller	PowerShell

Group Policy

Policy	What it does	Reason
Minimum Password length	Password has to be at least 12 characters	Make passwords harder to guest
Password Complexity	Passwords have to at least have 1 Capital and lowercase letter plus 1 Special character	Enforce stronger passwords
Screen lock	Lock the user out if no activity for 15 min (900 seconds)	Keep unattended machines safe
Removable storage Access	Make sure user cannot get access to the computer using external sources	Prevents USB data exfiltration

Common Help desk Task

- Password reset with forced password change on next login and account unlock

- Disable account after offboarding
- Inactive account audit
- Group membership reporting

Key PowerShell Commands Used

- `Install-WindowsFeature -Name GPMC (Group policies Management Console)`
- `New-ADOrganizationalUnit -Name " " -Path "DC= ,DC= "`
- `New-ADGroup -Name " " -GroupScope Global -GroupCategory Security -Path "OU= ,DC= ,DC= "`
- `$password = ConvertTo-SecureString "Password@2026!" -AsPlainText -Force`
- `Add-ADGroupMember -Identity "HR Group" -Members "Magic.Miller"`
- `Set-ADAccountPassword -Identity "Magic.Miller" -Reset -NewPassword (ConvertTo-SecureString "NewPass@2026!" -AsPlainText -Force)`
- `Set-ADUser -Identity "Magic.Miller" -ChangePasswordAtLogon $true`
- `Unlock-ADAccount -Identity "Magic.Miller"`
- `Disable-ADAccount -Identity "Magic.Miller"`
- `Search-ADAccount -AccountDisabled | Select-Object Name, SamAccountName`
- `Get-ADUser -Filter {LastLogonDate -lt '$cutoff' -and Enabled -eq $true} -Properties LastLogonDate | Select-Object Name, LastLogonDate`
- `Get-ADPrincipalGroupMembership -Identity "Magic.Miller" | Select-Object Name`
- `New-ADUser -Name "Magic.Miller" -GivenName "Magic" -Surname "Miller" ` - SamAccountName "Magic.Miller" -UserPrincipalName "Magic.Miller@Test.Forest" ` -Path "OU=HR,DC=Test,DC=Forest" -AccountPassword $password -Enabled $true`
- `$cutoff = (Get-Date).AddDays(-90)`
`Get-ADUser -Filter {LastLogonDate -lt $cutoff -and Enabled -eq $true} ` -Properties LastLogonDate | Select-Object Name`